

L05 Potential and Congestion Games

CS 280 Algorithmic Game Theory

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Potential Games

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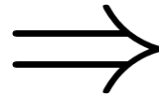
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Example (Battle of the sexes).

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Row 1	5, 2	-1, -2
Row 2	-5, -4	1, 4



Action	Col 1	Col 2
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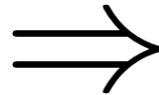
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Weighted Potential Games:

where $w_i > 0$.

$$\Phi(s_i, s_{-i}) - \Phi(s'_i, s_{-i}) = w_i \cdot (u_i(s_i, s_{-i}) - u_i(s'_i, s_{-i})),$$

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$$\Phi(s_i^*, s_{-i}^*) - \Phi(s'_i, s_{-i}^*) = u_i(s_i^*, s_{-i}^*) - u_i(s'_i, s_{-i}^*) < 0.$$

Contradiction!

Potential Games

Algorithm (Greedy).

1. Initialize $s^{(0)}$ arbitrarily and set $t = 0$.
2. **Loop**
3. Find player i and strategy s'_i with $u_i(s'_i, s_{-i}^{(t)}) > u_i(s^{(t)})$.
4. If no such pair exists, **return** $s^{(t)}$.
5. Set $s^{(t+1)} = (s'_i, s_{-i}^{(t)})$.
6. Set $t = t + 1$ and repeat.

Lemma. *The algorithm above reaches a pure Nash equilibrium.*

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- A run ends at a sink, hence at a pure Nash equilibrium.

Congestion Games

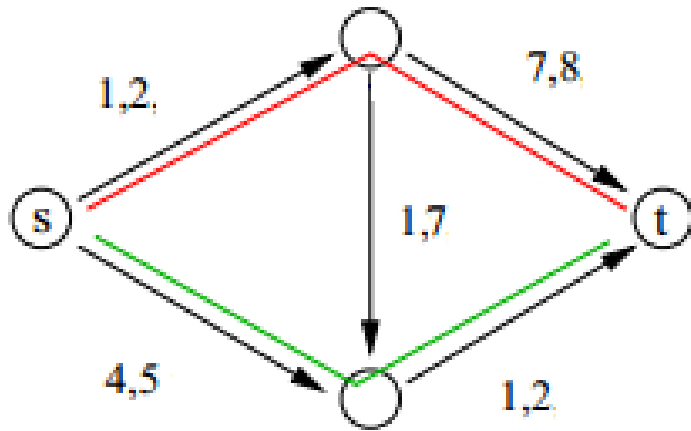
A **congestion game** is defined by:

- A set of players $[n] = \{1, \dots, n\}$.
- A finite set E of edges, facilities, or bins.
- A nonempty strategy set $S_i \subseteq 2^E$ for each player i .
- A cost function $c_e : \{1, \dots, n\} \rightarrow \mathbb{R}_{\geq 0}$ for each $e \in E$.

For a profile $s = (s_1, \dots, s_n)$:

- $l_e(s) = |\{i : e \in s_i\}|$ is the load on edge e .
- Player i minimizes $c_i(s) = \sum_{e \in s_i} c_e(l_e(s))$.

Congestion Games



For this game:

$n = 2$, with players 1 and 2.

E : the network's directed edges.

S_i : all s - t paths.

Each edge lists $c_e(1), c_e(2)$.

Player 1: upper path. Player 2: lower path.

Remark: Rosenthal introduced these games in 1973.

They model atomic routing games.

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Fix player i and profiles s, s' that differ only in player i 's strategy.

- $\Phi(s) = \sum_{e \in s_i \cap s'_i} \sum_{j=1}^{l_e(s)} c_e(j) + \sum_{e \in s_i \setminus s'_i} \sum_{j=1}^{l_e(s)} c_e(j) + \sum_{e \in s'_i \setminus s_i} \sum_{j=1}^{l_e(s)} c_e(j) + \dots$
- $\Phi(s') = \sum_{e \in s_i \cap s'_i} \sum_{j=1}^{l_e(s')} c_e(j) + \sum_{e \in s_i \setminus s'_i} \sum_{j=1}^{l_e(s')} c_e(j) + \sum_{e \in s'_i \setminus s_i} \sum_{j=1}^{l_e(s')} c_e(j) + \dots$

Remaining terms:

$$+ \sum_{e \in E \setminus (s_i \cup s'_i)} \sum_{j=1}^{l_e(s)} c_e(j) \qquad + \sum_{e \in E \setminus (s_i \cup s'_i)} \sum_{j=1}^{l_e(s')} c_e(j)$$

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Same load
 $l_e(s) = l_e(s') + 1$
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Remaining terms (unchanged loads):

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$$\Phi(s) - \Phi(s') = \sum_{e \in s_i \setminus s'_i} c_e(l_e(s)) - \sum_{e \in s'_i \setminus s_i} c_e(l_e(s'))$$

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Proof cont.

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Monderer–Shapley: every finite exact potential game has a congestion-game representation (allowing real resource costs).

An Algorithm for symmetric network congestion games

Assumption: All players have the same endpoints S and T (and thus they all have the same set of paths/strategies).

Basic idea: Min-cost flow reduction

Costs are nonnegative and nondecreasing in the load.

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Definition (Min-cost flow). In directed $G = (V, E)$, send demand $d \geq 0$ from s to t . Edge (u, v) has capacity $c(u, v) \geq 0$ and unit cost $a(u, v)$.

$$\min \sum_{(u,v) \in E} a(u, v) f(u, v)$$

$$0 \leq f(u, v) \leq c(u, v) \quad \forall (u, v) \in E \quad \text{capacity constraints}$$

$$\sum_{w: (u,w) \in E} f(u, w) = \sum_{w: (w,u) \in E} f(w, u) \quad \forall u \notin \{s, t\} \quad \text{flow conservation}$$

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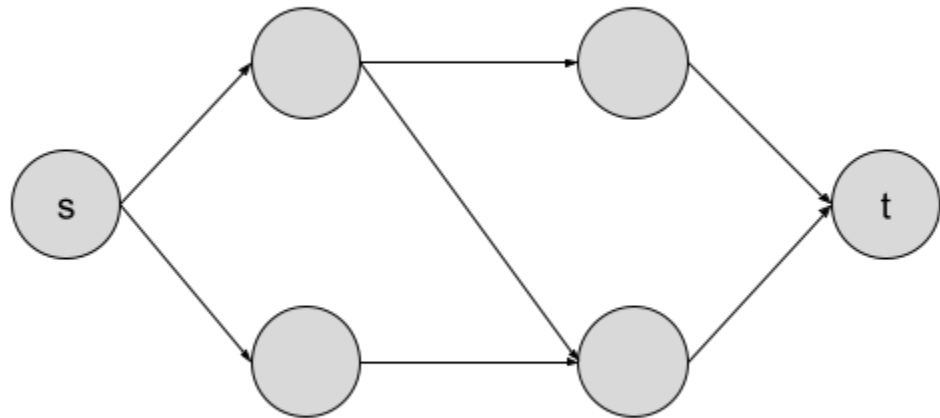
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Min-cost flow via LP!

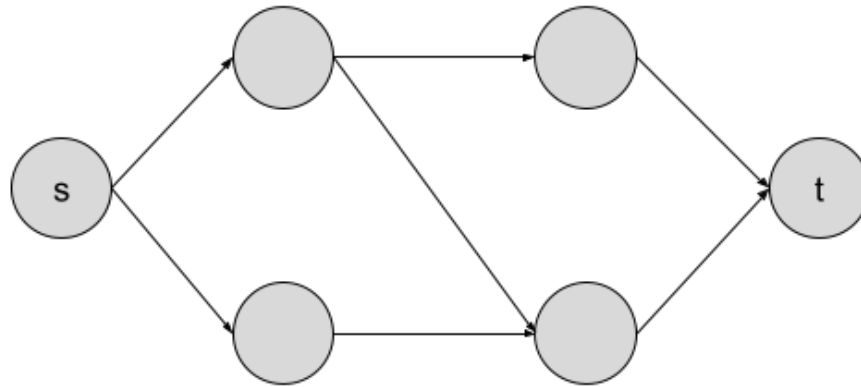
An Algorithm for symmetric network congestion games; the reduction

Initial graph in the Congestion Game.

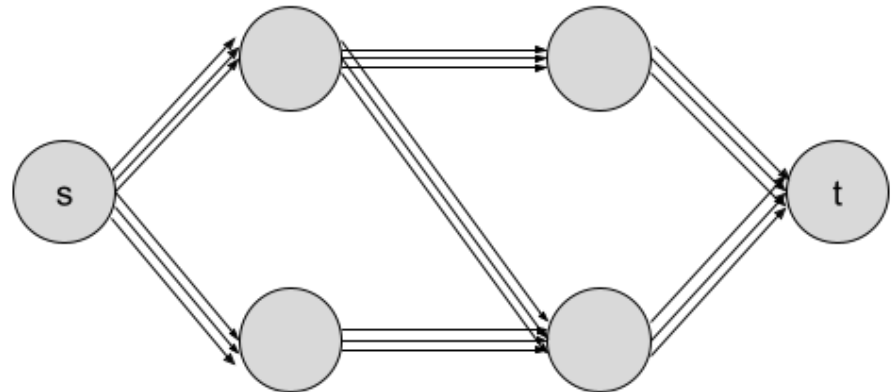


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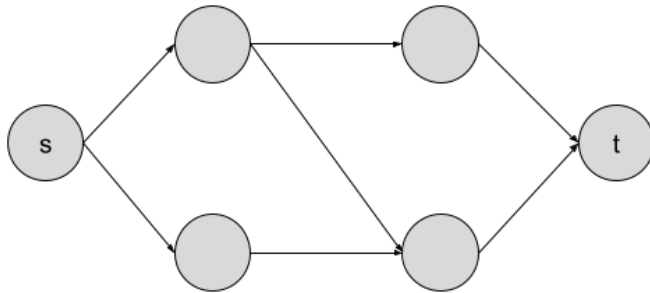
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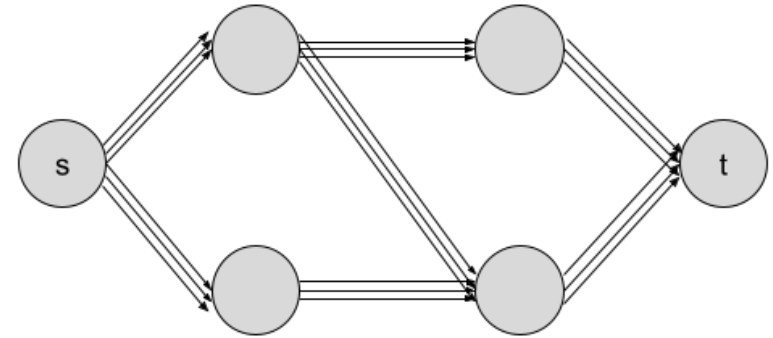
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Remove cycles, then decompose the integral flow into n paths.

An integral min-cost flow of value n yields a profile minimizing Φ , hence a pure Nash equilibrium.
HW1: justify the correspondence.